SMART WATER MONITORING SYSTEM

Early detection of drain overflow and underground pipeline leakage

1. PROBLEM & SOLUTION

PROBLEM

• Heavy rain can make drains overflow, causing waterlogging, traffic disruption and road damage.
• Underground pipe leaks waste water and may weaken surrounding soil or roads.
• These problems are often noticed only after visible damage occurs.

SOLUTION

• Place water-level, flow and pressure sensors at important points.
• A microcontroller continuously checks the readings against safe limits.
• When an abnormal condition is detected, the system gives an immediate warning.

SENSORS & COMPONENTS

• Ultrasonic water-level sensor — measures the water level in the drain.
• Flow sensor — measures water movement through the pipe.
• Pressure sensor — detects unusual pressure changes.
• Arduino/ESP32, buzzer, LEDs, display and power supply.

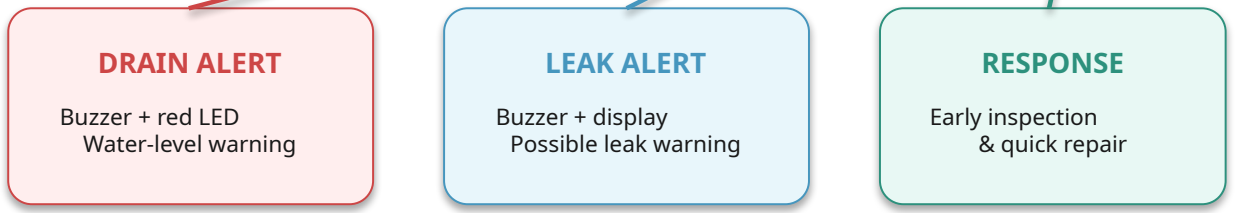
TECHNICAL WORKING

• Drain: the ultrasonic sensor sends a pulse and measures its echo time. As water rises, the distance to the water surface decreases. If it crosses the critical level, an alert is triggered.
• Pipeline: flow and pressure are compared with normal values. A sudden pressure drop or unexpected flow difference can indicate a possible leak.

2. SMART DETECTION & ALERT SYSTEM



3. OUTPUT / ACTION



MODEL PREVIEW • MINI ROAD + DRAIN + TRANSPARENT PIPE + PUMP + CONTROLLED LEAK